Kaalchakra

Every period, in its place. — Plan summary for PRD

Punjab Jobs AI Bootcamp · J-PAL South Asia · Hackathon 1 (Scheduling Optimizer) · individual · 120 minutes

1. Problem & goal

Colleges build timetables by hand each semester. Departments fit sections, labs and teachers into the week and compete for shared spaces (seminar halls, computer labs). It takes weeks and still produces clashes, uneven teaching loads and idle rooms. **Goal:** a web app that produces clash-free timetables for **any** college, used by everyone who depends on the timetable.

Judging criterion	How Kaalchakra wins it
Core – no clashes	Constraint solver + independent validator; “0 clashes – verified” seal; unplaced-with-reason instead of ever producing a clash
Usability	Multi-college platform, 5 role portals, Structure Builder (no hard-coded shape), one-click demo colleges, paste-from-Excel, plain-language errors, chatbot
Design	Editorial “Paper & Ink” theme (not a blue/green dashboard), GSAP motion, notice-board print
Advanced	Lab double periods, availability, cross-dept room exchange, lock & regenerate, absence re-plan, substitutes
Relevance	Shared-room fairness, teacher load balance, idle-room heatmap, leave flow, student Now/Next, student-teacher meeting requests

2. Multi-college platform — nothing hard-coded

One platform hosts many colleges; every record belongs to a college and no data crosses colleges. The number and names of departments, programmes/years, sections, students per section, rooms, teachers, working days, periods per day, timings, lunch and lab length, and the subject list of each programme all come from each college's configuration. Demo data is loaded through the same path: two differently shaped demo colleges (Engineering: 3 depts, 7 slots a day, Mon–Sat, staggered lunch; Arts & Science: 2 depts, 6 slots a day, Mon–Fri, 30-minute lunch) both produce clash-free timetables.

3. Users & roles (college- and department-scoped)

Role	Sees	Can do
Super Admin (platform / university body)	All colleges with status cards: counts, published?, clash seal, quality score	Create / edit / archive / clone colleges; Structure Builder: calendar, departments, programmes/years, number of sections, students per section; assign College Admins. No direct timetable edits
College Admin (Principal / Timetable office)	Own college: all timetables, dashboard, shared-room ledger, room heatmap, Conflict Center, all leave	Fine-tune structure, shared rooms, generate whole college, lock slots, resolve cross-dept clashes, publish, export
Department Admin (HOD)	Own dept sections, teachers, rooms, timetables, load chart, share of shared rooms, dept leave	CRUD own dept data, dept generate (against global ledger), mark Draft → Ready, Room Exchange, approve leave + pick substitute
Teacher	My Week, Today (now/next), weekly load, substitution duties, own-dept timetables (read), meeting requests inbox	Request leave, request unavailable slots, set office hours, accept / propose / decline meetings, print
Student	Own section's published timetable, Now/Next card, substitute notices, meeting status (mobile-first)	Pick college + section, request a meeting with a teacher, print, share link

Teachers and students read only the **published** version. All scoping goes through one access helper. Until authentication (built last) the session is chosen via a role switcher.

4. Semester timetable and leave cover

- **Whole semester, real dates:** each college sets its semester (e.g. 20 Jul – 27 Nov), holidays and exam/no-class days. The solver builds one clash-free weekly pattern and Kaalchakra rolls it out over every teaching date, skipping holidays, and reports contact hours delivered per subject.
- **Leave is for a date:** when the HOD approves it, the best **free** teacher is auto-allotted to each affected class in the same slot (same department → teaches the subject → knows the section → lightest load), never breaking limits or lunch. The HOD can swap the suggestion; if nobody is free the class is flagged uncovered. The weekly pattern and all

other dates stay unchanged.

5. Meet-a-Teacher (student and teacher)

- Student picks a teacher of their section/dept → **“Can I meet?”** shows slots where both the teacher and the section are free (office hours highlighted) → sends a request with topic.
- Teacher inbox: accept (optionally with a free room), propose another slot, or decline. Limit of pending requests per student is set per college.
- Meetings never block teaching; if a later re-plan moves a class into a meeting slot, the meeting is flagged “needs reschedule”. HOD sees counts only.

6. Cross-department rooms & clash prevention

- **Room ownership:** each room belongs to a department or is college-shared. A section may use a room only if its dept owns it, it is shared, or an approved grant covers that slot.
- **Room Exchange:** owner HOD offers idle rooms/slots; another HOD requests; approval creates a grant. Revoking flags affected classes for re-plan.
- **Global booking ledger:** rooms and teachers are global resources. A HOD's dept generate treats other departments' committed classes as fixed, so two departments can never book the same room or shared teacher.
- **Conflict Center:** College Admin sees any cross-dept clash with one-click fixes; publishing is blocked while a clash exists.

Hard rules (never broken): nothing crosses colleges · teacher, section and room single-booking across all departments · room type & capacity · teacher unavailability & leave · lunch · labs as consecutive periods not crossing lunch · teacher max per day/week · locked slots fixed · room ownership/grants. **Soft goals (scored 0–100):** subject spread across days, max one period of a subject per day, few teacher/section gaps, balanced daily load, own rooms before borrowed, no last-period labs.

7. Finding things & Kaal Assistant

- **Filters** on every timetable (dept, section, teacher, room, room type, subject, day, theory/lab) kept in the URL; **Ctrl+K** search jumps to any timetable.
- **Who's free?** pick day + period → free teachers (for substitutes, extra classes) and free rooms (type, capacity, borrowable?). **Is X free?** week strip + next free slot. **Common free slot** for several teachers/rooms.
- **Kaal Assistant:** rule-based chatbot (offline, no API key) for small queries — “Is Prof. Kaur free Tue P3?”, “Where is CSE-3A now?”, “Free labs Friday after lunch?”, “Who is substituting today?”, “Why is ME-5B lab unplaced?”, “Can I meet Prof. Kaur tomorrow?”. Answers respect the user's role.

8. Design direction

“Paper & Ink” academic-register look: warm paper #F4EFE6, ink #1C1B19, vermilion #D9481C, marigold #E8A33D, plum #5B3A5E; dark “Blackboard” mode. Fraunces serif headings with a restrained type scale (hero 36–56px, body 15px), Inter UI, JetBrains Mono times. Subject cells in muted pastels, labs striped. GSAP: headline reveal, rotating Kaalchakra wheel while solving, cells drop into the grid, Flip animation on re-plan; disabled for reduced motion. Works at phone width and prints one timetable per A4.

9. Release roadmap (each version shippable, tested, committed, deployed)

Ver.	Min	Scope
v0.1	0–10	Scaffold, theme & fonts, multi-college data model, structure generator, 2 demo colleges built from config, GitHub + Vercel live
v0.2	10–30	Engine: solver, independent validator, semester calendar & delivered hours, leave cover (auto substitute), quality score, pre-flight check, unplaced reasons, Web Worker
v0.3	30–42	Timetable views by section/teacher/room, clash seal, “why here?” popover, GSAP, print
v0.35	42–50	Finder & filters: who's free, is X free, common free slot, filter bar, Ctrl+K
v0.4	50–62	Super Admin (colleges, Structure Builder) + College Admin portal: role & college switcher, setup, paste-from-Excel, generate all, publish
v0.5	62–72	HOD portal + Room Exchange: dept data, dept generate vs global ledger, Draft → Ready, grants
v0.6	72–84	Teacher & Student portals + Meet-a-Teacher: My Week, Today, leave, office hours, inbox; section view, Now/Next, substitute notice, meeting requests, share link

Ver.	Min	Scope
v0.7	84–94	Ops: leave inbox & substitute review screens, lock & regenerate, Conflict Center, dashboard, shared-room ledger
v0.8	94–102	Kaal Assistant chatbot (rule-based, role-scoped)
v0.9	102–110	Authentication: login, admin-managed users, salted hashed passwords, route guards, demo accounts
v1.0	110–120	All tests green, build, manual checklist on live URL, README, submit

If behind schedule, drop in order: share link → Ctrl+K → ledger → lock & regenerate → dark mode → auth. Solver tests are never dropped.

10. Quality & testing

- Unit tests (vitest) for every hard rule, impossible inputs, determinism, scoring, finder, access scopes, grants, re-plan, publishing, chatbot intents and auth — tracked in docs/TEST.md.
- Manual checklist on the live URL: demo flow per role, wizard from scratch, room exchange, leave → substitute, print, phone width, reduced motion, refresh keeps data.
- Docs: global CLAUDE.md (unchanging rules), docs/CLAUDE.md (design written before each version), docs/PROGRESS.md (build log → README), docs/TEST.md.

11. Out of scope (future work)

Server backend & real authentication (Supabase/Firebase, multi-tenant per college with row-level security), cross-college shared faculty, LLM-powered assistant, drag-and-drop editing, comparing multiple candidate timetables, exam timetables, notifications.

Tech: React 19 + Vite + TypeScript, Tailwind v4, zustand (localStorage), GSAP, papaparse, lz-string, lucide-react, vitest; fully client-side; deployed on Vercel.